DL ACTIVITY - Movie Review Sentiment Analysis

DATASET LINK :-
https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews

APP LINK :- https://deep-learning-activity-cpkfarvuegwzcja7ml5dco.streamlit.app/

GITHUB LINK :- https://github.com/anula-codes/Deep-learning-activity

Aim

Classify movie reviews as positive or negative with three deep learning models (LSTM, stacked BiLSTM, 1D CNN), compare them, and serve the best one in a Streamlit app.

Dataset and setup

- 50,000 reviews: 25,000 positive, 25,000 negative
- Split: 40,000 train, 10,Dataset and setup000 test; 20% of the training data held out for validation
- Tokenizer: 5,000-word vocabulary, reviews padded to 200 tokens

Models

Model	Architecture	Epochs	Batch size
1: LSTM	Embedding(128), LSTM(128), Dense(1)	5	64
2: Stacked BiLSTM	Embedding(64), BiLSTM(64), BiLSTM(32), Dense(32), Dense(1), dropout 0.3	8	64
3: 1D CNN	Embedding(128), Conv1D(128), Conv1D(64), Dense(64), Dense(32), Dense(1), dropout 0.5	10	128

Training

All models used the Adam optimizer, binary cross-entropy loss and a 20% validation split.

```
Model: "sequential"
```

Layer (type)	Output Shape	Param #
embedding (Embedding)	(None, 200, 128)	640,000
lstm (LSTM)	(None, 128)	131,584
dense (Dense)	(None, 1)	129

```
 Total params: 771,713 (2.94 MB)
 Trainable params: 771,713 (2.94 MB)
 Non-trainable params: 0 (0.00 B)
Epoch 1/5
500/500 ━━━━━━━━━━━━━━━━━━━━ 155s 307ms/step - accuracy: 0.8110 - loss: 0.4116 - val_accuracy: 0.8608 - val_loss: 0.3301
Epoch 2/5
500/500 ━━━━━━━━━━━━━━━━━━━━ 164s 328ms/step - accuracy: 0.8837 - loss: 0.2878 - val_accuracy: 0.8796 - val_loss: 0.2920
Epoch 3/5
500/500 ━━━━━━━━━━━━━━━━━━━━ 155s 311ms/step - accuracy: 0.9069 - loss: 0.2364 - val_accuracy: 0.8741 - val_loss: 0.3017
Epoch 4/5
500/500 ━━━━━━━━━━━━━━━━━━━━ 156s 311ms/step - accuracy: 0.9218 - loss: 0.2011 - val_accuracy: 0.8727 - val_loss: 0.3137
Epoch 5/5
500/500 ━━━━━━━━━━━━━━━━━━━━ 153s 307ms/step - accuracy: 0.9334 - loss: 0.1775 - val_accuracy: 0.8627 - val_loss: 0.3496

Model 1: LSTM -> Test accuracy: 0.8623 | Test loss: 0.3307 | Time: 13.9 min
```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
embedding_1 (Embedding)	(None, 200, 64)	320,000
bidirectional (Bidirectional)	(None, 200, 128)	66,048
dropout (Dropout)	(None, 200, 128)	0
bidirectional_1 (Bidirectional)	(None, 64)	41,216
dropout_1 (Dropout)	(None, 64)	0
dense_1 (Dense)	(None, 32)	2,080
dropout_2 (Dropout)	(None, 32)	0
dense_2 (Dense)	(None, 1)	33

Total params: 429,377 (1.64 MB)
Trainable params: 429,377 (1.64 MB)
Non-trainable params: 0 (0.00 B)

```
Epoch 1/8
500/500 ━━━━━━━━━━━━━━━━━━━━ 145s 282ms/step - accuracy: 0.8059 - loss: 0.4170 - val_accuracy: 0.8712 - val_loss: 0.2997
Epoch 2/8
500/500 ━━━━━━━━━━━━━━━━━━━━ 139s 276ms/step - accuracy: 0.8936 - loss: 0.2713 - val_accuracy: 0.8670 - val_loss: 0.3131
Epoch 3/8
500/500 ━━━━━━━━━━━━━━━━━━━━ 141s 282ms/step - accuracy: 0.9168 - loss: 0.2160 - val_accuracy: 0.8700 - val_loss: 0.3075
Epoch 4/8
500/500 ━━━━━━━━━━━━━━━━━━━━ 138s 274ms/step - accuracy: 0.9298 - loss: 0.1871 - val_accuracy: 0.8645 - val_loss: 0.3544
Epoch 5/8
500/500 ━━━━━━━━━━━━━━━━━━━━ 138s 275ms/step - accuracy: 0.9445 - loss: 0.1538 - val_accuracy: 0.8736 - val_loss: 0.3794
Epoch 6/8
500/500 ━━━━━━━━━━━━━━━━━━━━ 139s 278ms/step - accuracy: 0.9534 - loss: 0.1303 - val_accuracy: 0.8709 - val_loss: 0.4226
Epoch 7/8
500/500 ━━━━━━━━━━━━━━━━━━━━ 143s 279ms/step - accuracy: 0.9646 - loss: 0.1047 - val_accuracy: 0.8656 - val_loss: 0.4329
Epoch 8/8
500/500 ━━━━━━━━━━━━━━━━━━━━ 137s 274ms/step - accuracy: 0.9686 - loss: 0.0930 - val_accuracy: 0.8601 - val_loss: 0.4834

Model 2: Stacked BiLSTM -> Test accuracy: 0.8596 | Test loss: 0.4803 | Time: 18.7 min
```

Model: "sequential_2"

Layer (type)	Output Shape	Param #
embedding_2 (Embedding)	(None, 200, 128)	640,000
conv1d (Conv1D)	(None, 196, 128)	82,048
max_pooling1d (MaxPooling1D)	(None, 98, 128)	0
conv1d_1 (Conv1D)	(None, 94, 64)	41,024
global_max_pooling1d (GlobalMaxPooling1D)	(None, 64)	0
dense_3 (Dense)	(None, 64)	4,160
dropout_3 (Dropout)	(None, 64)	0
dense_4 (Dense)	(None, 32)	2,080
dense_5 (Dense)	(None, 1)	33

Total params: 769,345 (2.93 MB)
Trainable params: 769,345 (2.93 MB)
Non-trainable params: 0 (0.00 B)

```
Epoch 1/10
250/250 ━━━━━━━━━━━━━━━━━━━━ 75s 294ms/step - accuracy: 0.7171 - loss: 0.5141 - val_accuracy: 0.8585 - val_loss: 0.3324
Epoch 2/10
250/250 ━━━━━━━━━━━━━━━━━━━━ 74s 295ms/step - accuracy: 0.8840 - loss: 0.2903 - val_accuracy: 0.8737 - val_loss: 0.2959
Epoch 3/10
250/250 ━━━━━━━━━━━━━━━━━━━━ 84s 302ms/step - accuracy: 0.9233 - loss: 0.2038 - val_accuracy: 0.8710 - val_loss: 0.3153
Epoch 4/10
250/250 ━━━━━━━━━━━━━━━━━━━━ 72s 287ms/step - accuracy: 0.9521 - loss: 0.1348 - val_accuracy: 0.8706 - val_loss: 0.3641
Epoch 5/10
250/250 ━━━━━━━━━━━━━━━━━━━━ 72s 290ms/step - accuracy: 0.9746 - loss: 0.0778 - val_accuracy: 0.8706 - val_loss: 0.4354
Epoch 6/10
250/250 ━━━━━━━━━━━━━━━━━━━━ 89s 320ms/step - accuracy: 0.9818 - loss: 0.0535 - val_accuracy: 0.8719 - val_loss: 0.4700
Epoch 7/10
250/250 ━━━━━━━━━━━━━━━━━━━━ 90s 354ms/step - accuracy: 0.9869 - loss: 0.0394 - val_accuracy: 0.8716 - val_loss: 0.5039
Epoch 8/10
250/250 ━━━━━━━━━━━━━━━━━━━━ 133s 317ms/step - accuracy: 0.9946 - loss: 0.0180 - val_accuracy: 0.8711 - val_loss: 0.6525
Epoch 9/10
250/250 ━━━━━━━━━━━━━━━━━━━━ 82s 330ms/step - accuracy: 0.9943 - loss: 0.0183 - val_accuracy: 0.8670 - val_loss: 0.6640
Epoch 10/10
250/250 ━━━━━━━━━━━━━━━━━━━━ 183s 493ms/step - accuracy: 0.9927 - loss: 0.0205 - val_accuracy: 0.8702 - val_loss: 0.7540

Model 3: 1D CNN -> Test accuracy: 0.8709 | Test loss: 0.7307 | Time: 15.9 min
```

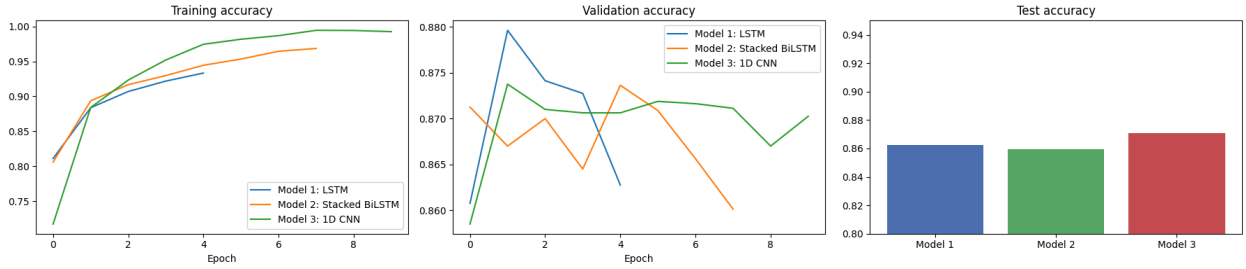
Results

Model 3 (1D CNN) had the highest test accuracy, about 0.87, but all three models landed between 0.86 and 0.87.

Model	Test accuracy
1: LSTM	about 0.86
2: Stacked BiLSTM	about 0.86
3: 1D CNN	about 0.87

Model 3 overfits: training accuracy reaches about 0.99 while validation stays near 0.87.

	Hidden layers	Epochs	Params	Train acc	Val acc	Test acc	Test loss	Time (s)
Model 1: LSTM	2.0	5.0	771713.0	0.9334	0.8627	0.8623	0.3307	832.0
Model 2: Stacked BiLSTM	3.0	8.0	429377.0	0.9686	0.8601	0.8596	0.4803	1124.0
Model 3: 1D CNN	4.0	10.0	769345.0	0.9927	0.8702	0.8709	0.7307	954.0



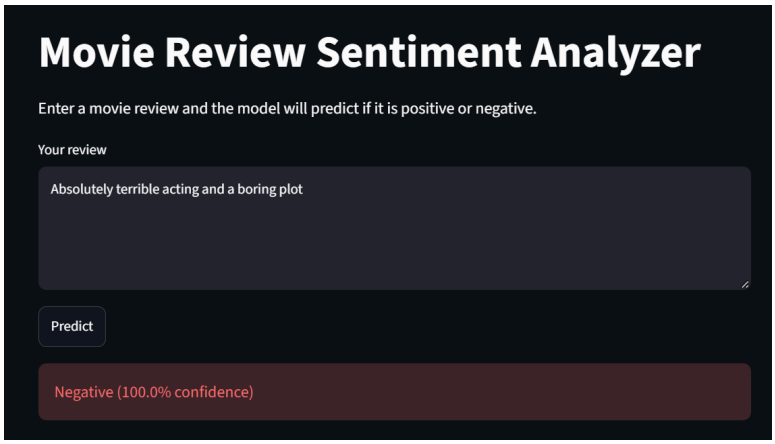
Sample predictions

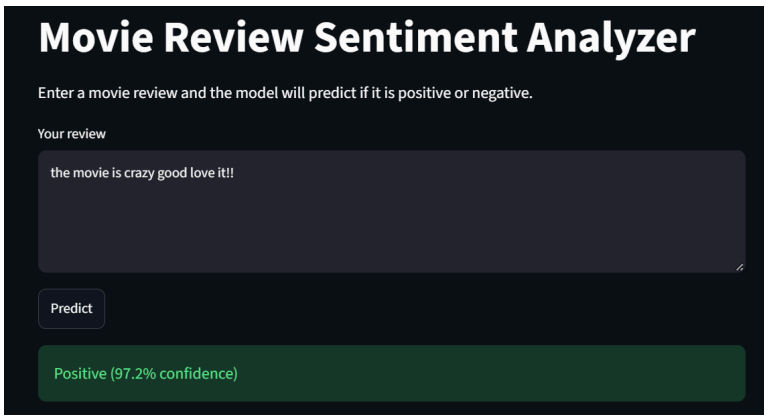
All three models gave the same label on the four sample reviews.

```
This movie was fantastic, I loved it
    Model 1: LSTM -> ('Positive', 0.958)
    Model 2: Stacked BiLSTM -> ('Positive', 0.966)
    Model 3: 1D CNN -> ('Positive', 1.0)
This movie was not that good
    Model 1: LSTM -> ('Negative', 0.359)
    Model 2: Stacked BiLSTM -> ('Negative', 0.216)
    Model 3: 1D CNN -> ('Negative', 0.37)
The movie was okay, but not that good
    Model 1: LSTM -> ('Negative', 0.267)
    Model 2: Stacked BiLSTM -> ('Negative', 0.267)
    Model 3: 1D CNN -> ('Negative', 0.025)
Absolutely terrible acting and a boring plot
    Model 1: LSTM -> ('Negative', 0.115)
    Model 2: Stacked BiLSTM -> ('Negative', 0.003)
    Model 3: 1D CNN -> ('Negative', 0.0)
```

Deployment

1. Saved the best model (best_model.keras) and the tokenizer (tokenizer.pkl) from Colab.
2. Built app.py with Streamlit: a text box and a Predict button.
3. Installed the requirements and ran the app with `streamlit run app.py`.





Live app link: https://deep-learning-activity-cpkfarvuegwzcja7ml5dco.streamlit.app/

Conclusion

The 1D CNN scored best on the test set (about 0.87), though the three models were close. It now powers a Streamlit app that labels any typed review as positive or negative. Early stopping or more dropout would reduce its overfitting.